

Log File Analyzer for Intrusion Detection

Introduction

Log files are very important in cybersecurity because they help track system activity and detect suspicious behavior. By analyzing login records, it is possible to identify repeated failed login attempts, suspicious IP addresses, and possible brute-force attacks. The aim of this project is to create a simple Log File Analyzer that checks login logs and identifies suspicious activity.

Abstract

This project is a Python-based Log File Analyzer developed to detect suspicious login attempts from sample log data. The program reads log entries, identifies failed login attempts, counts the number of failures from each IP address, and detects possible brute-force attacks when the same IP fails multiple times. It then displays a simple security report showing total failed attempts, suspicious IP activity, and alerts.

Tools Used

- Python
- Programiz Python Compiler / Python IDE
- GitHub

Steps Involved in Building the Project

1. Created sample login log data inside the Python program.
2. Read each log entry one by one.
3. Checked which log entries contained failed login attempts.
4. Counted the total number of failed logins.
5. Extracted IP addresses from failed login records.
6. Counted how many failed attempts came from each IP address.
7. Displayed suspicious IP activity in the output.
8. Added a simple brute-force detection rule to show an alert when an IP had 3 or more failed login attempts.

Conclusion

Through this project, I understood how log analysis is used in cybersecurity to monitor login activity and detect suspicious behavior. I also learned how to use Python concepts such as loops, conditions, string splitting, and dictionaries to analyze log data. This project helped me understand how failed login attempts can be tracked and how possible brute-force attacks can be identified in a simple way.